Taejoong (Tijay) Chung

Associate Professor (tijay@vt.edu, https://taejoong.github.io)

Education

09/2009–02/2015 **Ph.D.**, Computer Science and Engineering, Seoul National University, Seoul, Korea.
- Supervisors: Prof. Yanghee Choi and Prof. Ted "Taekyoung" Kwon

03/2005–09/2009 **B.S.**, Computer Science and Engineering, POSTECH (Pohang University of Science and Technology), Pohang, Korea.

Professional Experience

07/2025–present **Associate Professor**, Department of Computer Science, Virginia Tech, Blacksburg, VA, United States.

08/2020–06/2025 **Assistant Professor**, Department of Computer Science, Virginia Tech, Blacksburg, VA, United States.

02/2023–03/2026 **Adjunct Professor**, Department of Computer Science, POSTECH, Pohang, South Korea.

08/2018–07/2020 **Assistant Professor**, Department of Computer Science, Rochester Institute of Technology (RIT), Rochester, NY, United States.

09/2015–08/2018 **Postdoctoral Researcher**, Northeastern University, Boston, MA, United States.

Honors and Awards

2026 IETF/IRTF Applied Networking Research Prize (ANRP)

2024 National Science Foundation (NSF) CAREER Award

2024 Outstanding New Assistant Professor Award at the College of Engineering, Virginia Tech

2023 Mutually Agreed Norms for Routing Security (MANRS) Mentor

2023, 2025 College of Engineering Dean's Memorandum for Outstanding Teaching Evaluations

2022 Best Paper Honorable Mention Award, ACM CCS'22

2022 Mutually Agreed Norms for Routing Security (MANRS) Research Fellow

2019 Distinguished Paper Award, ACM IMC'19

2019 IETF/IRTF Applied Networking Research Prize (ANRP), IETF'105

2019 National Science Foundation (NSF) CRII Award

2017 Distinguished Paper Award, USENIX Security'17

2015 Microsoft Azure Research "Award Program: Microsoft Azure for Research Supports $20,000"

Grants

2026 RPKI Operational Health Dashboard for the ARIN Service Region
- American Registry for Internet Numbers (ARIN) Community Grant Program, 09/2026 – 10/2027, Sole Primary Investigator, $17,350

2026 Verified Agent Identity for Internet-Scale AI Ecosystems: A Measurement-Driven Defense for the MCP and A2A Trust Anchors
- Commonwealth Cyber Initiative, 07/2026 – 06/2027, Sole Primary Investigator, $100,000

2026 Measuring and Mitigating Motion Telemetry Leakage in Networked XR
- Commonwealth Cyber Initiative, 07/2026 – 06/2027, Co-Primary Investigator, joint with Bo Ji (VT), Brendan David-John (VT), $35,000

2025 Towards a Comprehensive View of the ROV Ecosystem: Unraveling Relying Parties, ROV Deployments, and Global Routing Implications

- Comcast Innovation Fund, 08/2025 – 07/2026, Sole Primary Investigator, $50,000

2024 Holistic Measurement and Mitigation of Email Security Vulnerabilities: Infrastructure, Tools, and Commercialization

- Commonwealth Cyber Initiative, 10/2024– 10/2025, Sole Primary Investigator, $50,000

2024 Leveraging ML for Advanced IRR Validation: Towards a More Secure Internet Infrastructure

- Comcast Innovation Fund, 08/2024 – 07/2025, Sole Primary Investigator, $30,000

2024 CAREER: Securing and Evolving Internet Security Protocols for Naming and Routing

- National Science Foundation. CNS-2339378, 05/2024 – 04/2029. Sole Primary Investigator $691,258 (VT), $691,258 (total)

2023 RoVISTA: Understanding the Current Validation Status of RPKI

- Google, Unrestricted Gift. 09/2023. Sole Primary Investigator $30,000 (VT)

2023 IMR: MT: Tools for Measuring Route Origin Validation in Resource Public Key Infrastructure (RPKI) at Scale

- National Science Foundation. CNS-2323137, 12/2023 – 11/2025. Sole Primary Investigator $599,997 (VT), $599,997 (total)

2023 SaTC: CORE: Medium: Cryptographic accumulators and revocation of credentials

- National Science Foundation. CNS-2247306, 05/2023 – 04/2026. Primary Investigator, joint with Foteini Baldimtsi (GMU), Anna Lysyanskaya (Brown), Charalampos Papamanthou (Yale), Alan Mislove (NEU). $240,000 (VT), $1,200,000 (total)

2023 A Study on Key Management Structure for Multi-Metaverse Avatar Authentication and Anonymization Assurance

- Ministry of Science and ICT, 01/2023 – 01/2026, Primary Investigator, $225,000 (VT), $163 M (total)

2022 Measuring and Auditing Route Origin Validation (ROV) of Network Operators for RPKI

- Comcast Innovation Fund, 02/2022 – 05/2023, Sole Primary Investigator, $75,000

2022 Rapidly Bootstrapping Implementation of DNS-Based Security Extensions with AI-driven Approach

- 4-VA, 06/2022 – 05/2023, Sole Primary Investigator, $25,000

2022 Improving Routing Security with a Complete View of Routing Origin Validation (ROV) in RPKI

- Commonwealth Cyber Initiative, 05/2022 – 08/2022, Sole Primary Investigator, $15,000

2021 Enhancing the Privacy and Reliability of Massive-scale Bluetooth Low Energy Contact Tracing

- Commonwealth Cyber Initiative, 06/2021 – 05/2022. Co-Primary Investigator, joint with Daphne Yao (VT), Carol Fung (VCU) $100,000 (VT), $150,000 (total)

2019 CNS Core: Large: Collaborative Research: Towards an Evolvable Public Key Infrastructure

- National Science Foundation. CNS-1901090, 06/2019 – 05/2022. Primary Investigator, joint with Dave Levin (UMD), Bruce M. Maggs (Duke University), Alan Mislove (NEU), Christo Wilson (NEU), Bryan Parno (CMU). $300,000 (VT), $2,000,000 (total)

2019 CRII: SaTC: Measuring and Improving the Management of Resource Public Key Infrastructure (RPKI)

- National Science Foundation. CNS-1850465, 10/2019 – 09/2021. Sole Primary Investigator. $166,561

2015 Personal Recommendation System and Privacy Protection through Big Data Analysis on Online Social Networks

- National Research Foundation of Korea (NRF). NRF-2016018668, 03/2015 – 02/2016. Sole Primary Investigator. ₩50,000,000 (≈ $44,000)

Refereed Conference Publications (My advisees are underlined)

NDSS'27 **pqRPKI: A Practical RPKI Architecture for the Post-Quantum Era**
W. Li, Y. Li, and T. Chung
ISOC NDSS, Seoul, Korea, February 2027

IMC'26 **A First Look at Certificate Revocation Latency**
N. Fooda, J. Larisch, J. M. Schanck, B. M. Maggs, T. Chung, and D. Levin
ACM Internet Measurement Conference (IMC'26), Karlsruhe, Germany, October 2026

IMC'26 **Split Alignment: Diffusing SPF Vulnerabilities With DMARC**
M. Hamza, M. Jonker, R. Sommese, S. Fernandez, O. Hureau, E. Pauley, and T. Chung
ACM Internet Measurement Conference (IMC'26), Karlsruhe, Germany, October 2026

IMC'26 **RScope: Unveiling Global ROV Deployments and Dependencies in the Post-ROV Era**
W. Li, Y. Xu, M. Zhang, V. Giotsas, and T. Chung
ACM Internet Measurement Conference (IMC'26), Karlsruhe, Germany, October 2026

SIGCOMM'26 **Comprehensive Revocation Checking at Scale: the Deployment of CRLite in Mozilla Firefox**
N. Fooda, J. Larisch, J. M. Schanck, T. Chung, D. Levin, B. M. Maggs, and C. Wilson
ACM SIGCOMM, Denver, United States, August 2026

USENIX Security'26 **iROV: Breaking the Silence of RPKI with Interactive Validation**
Y. Kim, S. Baek, W. Li, T. Chung, and M. S. Kang
USENIX Security, Baltimore, United States, August 2026

USENIX Security'26 **Motion in the Clear: Reconstructing VR User Behavior from Network Traffic**
J. Lee, J. Yoon, T. Chung, B. David-John, and B. Ji
USENIX Security, Baltimore, United States, August 2026

Oakland'26 **The Threat Landscape of IP Leasing in the RPKI Era**
W. Li, Y. Xu, and T. Chung
IEEE Symposium on Security and Privacy, San Francisco, United States, May 2026

NDSS'26 **Demystifying RPKI-Invalid Prefixes: Hidden Causes and Security Risks**
W. Li, T. Wan, and T. Chung
ISOC NDSS, San Diego, United States, February 2026

SIGCOMM'25 **Reliable and Decentralized Certificate Revocation via DNS: The Case for RevDNS**
P. Bhowmick, D. Levin, and T. Chung
ACM SIGCOMM, Coimbra, Portugal, September 2025

IMC'25 **Unraveling the Complexities of MTA-STS Deployment and Management in Securing Email**
Md. I. Ashiq, T. Fiebig, and T. Chung
ACM IMC, Madison, United States, October 2025

IMC'25 **Decoding DNSSEC Errors at Scale: An Automated DNSSEC Error Resolution Framework using Insights from DNSViz Logs**
Md. I. Ashiq, O. Hureau, C. Deccio, and T. Chung
ACM IMC, Madison, United States, October 2025

Oakland'25 **AccuRevoke: Enhancing Certificate Revocation with Distributed Cryptographic Accumulators**
M. Muid, T. Chung, and T. Hoang
IEEE Symposium on Security and Privacy, San Francisco, United States, May 2025

USENIX Security'25 **ImpROV: Measurement and Practical Mitigation of Collateral Damage of RPKI Route Origin Validation**
W. Li, Y. Li, and T. Chung
USENIX Security, Seattle, United States, August 2025

USENIX ATC'24 **mmTLS: Scaling the Performance of Encrypted Network Traffic Inspection**
J. Yoon, S. Do, D. Kim, T. Chung, and K. Park
USENIX Annual Technical Conference, Santa Clara, United States, July 2024

USENIX Security'24 **SPF Beyond the Standard: Management and Operational Challenges in Practice and Practical Recommendations**
Md. I. Ashiq, W. Li, T. Fiebig, and T. Chung
USENIX Security, Philadelphia, United States, August 2024

Oakland'24 **An Analysis of Recent Advances in Deepfake Image Detection in an Evolving Threat Landscape**
S. Muhammad Abdullah, A. Cheruvu, S. Kanchi, T. Chung, P. Gao, M. Jadliwala, and B. Viswanath
IEEE Symposium on Security and Privacy, San Francisco, United States, May 2024

NDSS'24 **IRRedicator: Pruning IRR with RPKI-Valid BGP Insights**
M. Kang, W. Li, R. van Rijswijk-Deij, T. Kwon, and T. Chung
ISOC NDSS, San Diego, USA, Feb. 2024

HotNets'23 **No Root Store Left Behind**
J. Larisch, W. Aqeel, T. Chung, E. Kohler, D. Levin, B. Maggs, B. Parno, and C. Wilson
ACM HotNets, Cambridge, USA, December 2023

ACSAC'23 **Delegation of TLS Authentication to CDNs using Revocable Delegated Credentials**
D. Yoon, T. Chung, and Y. Kim
ACSAC, Austin, United States, Canada, December 2023

IMC'23 **RoVista: Measuring and Understanding the Route Origin Validation (ROV) in RPKI**
W. Li, Z. Lin, Md. I. Ashiq, E. Aben, R. Fontugne, A. Phokeer, and T. Chung
ACM IMC, Montreal, Canada, October 2023

USENIX Security'23 **You've Got Report: Measurement and Security Implications of DMARC Reporting**
Md. I. Ashiq, W. Li, T. Fiebig, and T. Chung
USENIX Security, Anaheim, United States, August 2023

PAM'23 **Exploring the Evolution of the TLS Certificate Ecosystem**
S. Farhan and T. Chung
PAM, Virtual, March 2023

PAM'23 **Measuring TTL Violation of DNS Resolvers in the Wild**
P. Bhowmick, Md. I. Ashiq, C. Deccio, and T. Chung
PAM, Virtual, March 2023

IMC'22 **A Comparative Analysis of Certificate Pinning in Android & iOS**
A. Pradeep, M. Talha Paracha, P. Bhowmick, A. Davanian, A. Razaghpanah, T. Chung, M. Lindorfer, N. Vallina-Rodriguez, D. Levin, and D. Choffnes
ACM IMC, Nice, France, October 2022

CCS'22 **Hammurabi: A Framework for Pluggable, Logic-based X.509 Certificate Validation Policies**
J. Larisch, W. Aqeel, C. Wilson, A. Mislove, T. Chung, D. Levin, B. Parno, and B. Maggs
ACM CCS, Los Angeles, CA, USA, Nov 2022

USENIX Security'22 **Under the hood of DANE mismanagement in SMTP**
H. Lee, Md. I. Ashiq, M. Muller, R. van Rijswijk-Deij, T. Kwon, T. Chung
USENIX Security, Boston, United States, August 2022

SIGCOMM'21 **IP Unbound: Robust, Responsive CDN-scale service without CDN-Scale Addresses**
M. Fayed, V. Giotsas, L. Bauer, S. Kerola, M. Majkowski, P. Odintsov, J. Sitnicki, T. Chung, D. Levin, A. Mislove, N. Sullivan, C. Wood
ACM SIGCOMM, Virtual Conference, Aug. 2021

PAM'21 **Measurement and Analysis of Automated Certificate Reissuance**
O. Omolola, R. Roberts, I. Ashiq, T. Chung, D. Levin, and A. Mislove
PAM, Virtual Conference, Mar. 2021

IMC'20 **The Reality of Algorithm Agility: Studying the DNSSEC Algorithm Life-Cycle**
M. Mueller, W. Toorop, T. Chung, J. Janssen, and R. van Rijswijk-Deij
IMC, Pittsburgh, United States, Oct. 2020

FCS'20 **Assertion-Carrying Certificates**
W. Aqeel, Z. Hanif, J. Larisch, O. Omolola, T. Chung, D. Levin, B. Maggs, A. Mislove, B. Parno, and C. Wilson
Workshop on Foundations of Computer Security, Boston, United States, June 2020

USENIX Security'20 **A Longitudinal and Comprehensive Study of the DANE Ecosystem in Email**
H. Lee, A. Gireesh, R. van Rijswijk-Deij, T. Kwon, T. Chung
USENIX Security, Boston, United States, August 2020

CCS'19 **You Are Who You Appear to Be: A Longitudinal Study of Domain Impersonation in TLS Certificates**
R. Roberts, Y. Goldschlag, R. Walter, T. Chung, A. Mislove, D. Levin
ACM CCS, London, United Kingdom, Nov. 2019

IMC'19 **RPKI is Coming of Age: A Longitudinal Study of RPKI Deployment and Invalid Route Origins**
T. Chung, E. Aben, T. Bruijnzeels, B. Chandrasekaran, D. Choffnes, D. Levin, B. M. Maggs, A. Mislove, R. van Rijswijk-Deij, J. P. Rula, and N. Sullivan
ACM IMC, Amsterdam, The Netherlands, Oct. 2019

IMC'19 **Roll, Roll, Roll your Root: A Comprehensive Analysis of the First Ever DNSSEC Root KSK Rollover**
M. Müller, M. Thomas, D. Wessels, W. Hardaker, T. Chung, W. Toorop, and R. van Rijswijk-Deij
ACM IMC, Amsterdam, The Netherlands, Oct. 2019

NDSS'19 **maTLS: How to Make TLS middlebox-aware?**
H. Lee, Z. Smith, J. Lim, G. Choi, S. Chun, T. Chung, and T. Kwon
ISOC NDSS, San Diego, USA, Feb. 2019

IMC'18 **Is the Web Ready for OCSP Must Staple?**
T. Chung, J. Lok, B. Chandrasekaran, D. Choffnes, D. Levin, B. M. Maggs, A. Mislove, J. Rula, N. Sullivan, and C. Wilson
ACM IMC, Boston, USA, Nov. 2018

CSCW'18 **Privacy Leakage in Event-based Social Networks: A Meetup Case Study**
T. Chung, J. Han, D. Choi, T. Kwon, J. Rha, and H. Kim
ACM CSCW, Jersey City, USA, Nov. 2018

IMC'17 **Understanding the Role of Registrars in DNSSEC Deployment**
T. Chung, R. van Rijswijk-Deij, D. Choffnes, A. Mislove, C. Wilson, D. Levin, and B. M. Maggs
ACM IMC, London, UK, Nov. 2017

USENIX Security'17 **A Longitudinal, End-to-End View of the DNSSEC Ecosystem**
T. Chung, R. van Rijswijk-Deij, B. Chandrasekaran, D. Choffnes, D. Levin, B. M. Maggs, A. Mislove, and C. Wilson
USENIX Security, Vancouver, Canada, Aug. 2017

IMC'16 **Measuring and Applying Invalid SSL Certificates: The Silent Majority**
T. Chung, Y. Liu, D. Choffnes, D. Levin, B. M. Maggs, A. Mislove, and C. Wilson
ACM IMC, Santa Monica, CA, USA, Nov. 2016

IMC'16 **Tunneling for Transparency: A Large-Scale Analysis of End-to-End Violations in the Internet**
T. Chung, D. Choffnes, and A. Mislove
ACM IMC, Santa Monica, CA, USA, Nov. 2016

CCS'16 **Measurement and Analysis of Private Key Sharing in the HTTPS Ecosystem**
F. Cangialosi, T. Chung, D. Choffnes, D. Levin, B. M. Maggs, A. Mislove, and C. Wilson
ACM CCS, Vienna, Austria, Oct. 2016

COSN'15 **Characterizing Conversation Patterns in Reddit: From the Perspectives of Content Properties and User Participation Behaviors**
D. Choi, J. Han, T. Chung, Y. Ahn, B. Chun, and T. Kwon
ACM COSN, Stanford, CA, USA, Nov. 2015

COSN'15 **Sharing Topics in Pinterest: Understanding Content Creation and Diffusion Behaviors**
J. Han, D. Choi, A. Choi, T. Chung, T. Kwon, J. Rha, and C. Chuah
ACM COSN, Stanford, CA, USA, Nov. 2015

ANCS'14 **CoRC: Coordinated Routing and Caching for Named Data Networking**
H. Choi, J. Yoo, T. Chung, N. Choi, T. Kwon, and Y. Choi
IEEE ANCS, Marina del Rey, CA, USA, Oct. 2014

WWW'14 **Unveiling Group Characteristics in Online Social Games: A Socio-Economic Analysis**
T. Chung, J. Han, D. Choi, T. Kwon, H. K. Kim and Y. Choi
ISOC WWW, Seoul, Korea, Apr. 2014

IFIP Networking'13 **Spatial and Temporal Locality of Content in BitTorrent: A Measurement Study**
T. Chung, J. Han, H. Lee, J. Kangasharju, T. Kwon, and Y. Choi
IFIP Networking, Brooklyn, NY, USA, May 2013

SIGMETRICS'12 **Bundling Practice in Bittorrent: What, How, and Why**
J. Han, S. Kim, T. Chung, T. Kwon, H. Kim, and Y. Choi
ACM SIGMETRICS, London, UK, Jun. 2012

IFIP Networking'12 **Content Publishing and Downloading Practice in BitTorrent**
S. Kim, J. Han, T. Chung, T. Kwon, H. Kim, and Y. Choi
IFIP Networking, Prague, Czech, May 2012

GLOBECOM'12 **Bandwidth Allocation for BitTorrent under Multi-Torrent Environments**
J. Choi, J. Han, T. Chung, E. Cho, T. Kwon, and Y. Choi
IEEE GLOBECOM, Houston, Texas, United States, Dec. 2011

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Journal Publications

Privacy Guarantees of BLE Contact Tracing: A Case Study on COVIDWISE.
Salman Ahmed, Ya Xiao, T. Chung, Carol Fung, Moti Yung, and Danfeng (Daphne) Yao.
IEEE Computer Society, 2021

Rolling with Confidence: Managing the Complexity of DNSSEC Operations
Moritz Muller, T. Chung, Alan Mislove, and Roland van Rijswijk-Deij
IEEE Transactions on Network and Service Management (IEEE TNSM), 2019

Predicting Content Consumption from Content-to-Content Relationships
Jinyoung Han, Daejin Choi, T. Chung, Chen-Nee Chuah, Hyun-chul Kim, and Ted "Taekyoung" Kwon
Elsevier Journal of Network and Computer Applications (JNCA), Vol. 132, pp. 1, April 2019

A First Look at Certification Authority Authorization (CAA)
Quirin Scheitle, T. Chung, Jens Hiller, Oliver Gasser, Johannes Naab, Roland van Rijswijk-Deij, Oliver Hohlfeld, Ralph Holz, Dave Choffnes, Alan Mislove, and Georg Carle
ACM SIGCOMM Computer Communications Review (CCR), Vol. 48, pp. 2, Apr. 2018

An End-to-End View of DNSSEC Ecosystem Management
T. Chung, R. van Rijswijk-Deij, B. Chandrasekaran, D. Choffnes, D. Levin, B. M. Maggs, A. Mislove, and C. Wilson
USENIX Magazine ;login:, Vol. 42, No. 4, Dec. 2017

A Target-centric Surveillance System based on Localization and Social Networking
J. Han, N. Choi, T. Chung, T. Kwon, and Y. Choi
Multimedia Tools and Applications (MTAP), Vol. 73, Issue 1, pp. 241-265, Nov. 2014

Strategic Bundling for Content Availability and Fast Distribution in BitTorrent
J. Han, T. Chung, S. Kim, H. Kim, J. Kangasharju, T. Kwon, and Y. Choi
Computer Communication (COMCOM), Vol. 43, pp. 64-73, May 2014

Bundling Practice in Bittorrent: What, How, and Why
J. Han, S. Kim, T. Chung, T. Kwon, H. Kim, and Y. Choi
ACM SIGMETRICS Performance Evaluation Review, Vol. 40, No. 1, pp. 77-88, Jun. 2012.

How Prevalent is Content Bundling in BitTorrent?
J. Han, T. Chung, S. Kim, T. Kwon, H. Kim, and Y. Choi
ACM SIGMETRICS Performance Evaluation Review, Vol. 39, No. 1, pp. 319-320, Jun. 2011.

Posters

NSDI'23 **Scaling the Performance of Modern TLS Traffic Monitoring without Compromising the Security**
J. Yoon, S. Do, D. Kim, T. Chung, and K. Park
USENIX NSDI, Boston, USA, Apr. 2023

ASIACCS'22 **Privacy Guarantees of BLE Contact Tracing: A Case Study on COVIDWISE**
S. Ahmed, Y. Xiao, T. Chung, C. Fung, M. Yung, and D. Yao
ACM ASIACCS, Nagasaki, Japan, Jun. 2022

IMC'19 **Tracking Registrar Support for DNSSEC: It's Slowly Getting Better**
S. Roth, R. van Rijswijk-Deij, and T. Chung
ACM IMC, Amsterdam, The Netherlands, Oct. 2019

SIGCOMM'17 **The Root Canary: Monitoring and Measuring the DNSSEC Root Key Rollover**
R. van Rijswijk-Deij, T. Chung, D. Choffnes, A. Mislove, and W. Toorop
SIGCOMM, Los Angeles, CA, USA, Aug. 2017

ACNS'14 **Toward Terabyte-scale Caching with SSD in a Named Data Networking Router**
W. So, T. Chung, H. Yuan, D. Oran, and M. Stapp
ANCS, Marina del Rey, CA, USA, Oct. 2014

PAM'13 **Spatial and Temporal Locality of Swarm Dynamics in BitTorrent**
T. Chung, J. Han, H. Lee, J. Kangasharju, T. Kwon, and Y. Choi
PAM, Hong Kong, March. 2013.

Non-Refereed Conference Publications

NIST PQC'25 **Post-Quantum Diversity for DNSSEC: Routine Performance, Resilient Fallback**
T. Chung, J. Harvey, B. Kaliski, D. McVicker, B. Overeinder, S. Sheth, and O. Surý
NIST Post-Quantum Cryptography Standardization Conference, Maryland, USA, May 2025

TPRC'17 **Investigating End-To-End Integrity Violations in Internet Traffic**
A. Mislove, D. Choffnes, and T. Chung
TPRC, Arlington, VA, USA, Sept. 2017

Internet Drafts and Preprints

IETF I-D **Post-Quantum Authentication for the Resource Public Key Infrastructure**
W. Li and T. Chung
IETF SIDROPS Working Group, Internet Draft (draft-li-sidrops-pqrpki), Jul. 2026

arXiv **Enc53: DNSSEC-Anchored Stateless Tickets for Post-Quantum Authoritative DNS**
M. H. Tran, M. R. A. Muid, and T. Chung
arXiv:2609.14210, 2026

IETF I-D **Post-Quantum Cryptography Strategy for DNSSEC**
S. Sheth, T. Chung, and B. Overeinder
IETF, Internet Draft, Oct. 2025

arXiv **ASINT: Learning AS-to-Organization Mapping from Internet Metadata**
Y. Xu, W. Li, E. Umrani, and T. Chung
arXiv:2508.02571, 2025 (presented at NANOG 95)

IETF I-D **Research Agenda for a Post-Quantum DNSSEC**
A. M. Fregly, R. van Rijswijk-Deij, M. Müller, P. Thomassen, C. Schutijser, and T. Chung
IETF, Internet Draft, Sep. 2023

Teaching

F20, F23 **CS 6204**, Public Key Infrastructure and Network Security.

F22, F25 **CS 5584**, Network Security.

S24 **CS/ECE 5560**, Fundamentals of Information Security.

F23 **CS 4664**, Data-Centric Computing Capstone.

F23 **CS 4274**, Secure Computing Capstone.

S21, S22, S25 **CS 4264**, Principles of Computer Security.

F18, F19 **CSCI 351**, Data Communication and Networks.

Advising

01/2026 – Present **Weitong Li**, Postdoc, Virginia Tech.

08/2026 – Present **Mingxi Qiu**, Ph.D. Students, Virginia Tech.

08/2026 – Present **Sultanul Arifin**, Ph.D. Students, Virginia Tech.

02/2025 – Present **Yongzhe Xu**, Ph.D. Students, Virginia Tech.

01/2024 – Present **Hoang Tran**, Ph.D. Students, Virginia Tech, interned at Verisign.

08/2023 – Present **Munshi Rejwan Ala Muid**, Ph.D. Students, Virginia Tech, Co-advised with Thang Hoang; Bitshare Fellowship Recipient.

08/2024 – 07/2025 **Muhammad Hamza**, Ph.D. Students, Virginia Tech, now advised by Eric Pauley.

01/2021 – 12/2025 **Mohammad Ishtiaq Ashiq Khan**, Ph.D. Students, Virginia Tech, interned at Meta; now at Meta.

01/2021 – 12/2025 **Weitong Li**, Ph.D. Students, Virginia Tech, interned at Cloudflare and CableLabs; Pratt Fellowship Recipient; Graduate Student Award.

03/2025 – 08/2025 **Olivier Hureau**, Postdoc, Ph.D. from Université Grenoble Alpes, France, now Associate Professor (Maître de Conférences) at Université Grenoble Alpes.

– 2022 **Hyeonmin Lee**, Ph.D., Seoul National University (co-advised), now Assistant Professor at Hanyang University.

– 2024 **Mahesh Maddhuru**, M.Eng., Virginia Tech, at Akamai Technologies.

01/2021 – 11/2023 **Protick Bhowmick**, Masters, Virginia Tech, at Meta; interned at Cisco ThousandEyes.

08/2020 – 08/2022 **Syed Muhammad Farhan**, Masters, Virginia Tech, at Amazon.

2019 **Anna Kwon**, Research Intern, RIT, later at MIT.

2018 **Spencer Roth**, Research Intern, RIT, later at RIT.

2018 **Aniketh Girish**, Research Intern, RIT, later at IMDEA Networks.

Ph.D. Committees

Fall 2024 **Daegun Yoon**, KAIST.

Spring 2024 **Aravind Cheruvu**, Virginia Tech.

Spring 2024 **Thomas Nguyen**, Virginia Tech.

Spring 2024 **Zhongdong Liu**, Virginia Tech.

Spring 2024 **Miles Frantz**, Virginia Tech.

Spring 2024 **Muhammad Talha Paracha**, Northeastern University.

Spring 2023 **Tung Dao**, Virginia Tech.

Fall 2022 **Jiameng Pu**, Virginia Tech.

Service

Award Committee Member.

'21, '22, '23, '24, '25, '26 IETF/IRTF Applied Networking Research Prize (ANRP)

'21 ACM Student Research Competition (SRS) for SIGCOMM poster and demo session
General Chair.

'22 The ACM/IRTF Applied Networking Research Workshop (ANRW)
Program Committee Chair.

'26 The IFIP Network Traffic Measurement and Analysis Conference (TMA)
Steering Committee Member.

'23, '24 The ACM/IRTF Applied Networking Research Workshop (ANRW)
Program Committee Member.

'26 ACM SIGCOMM

'25, '26 USENIX Security (SEC)

'26 ACM Asia Computer and Communications Security (ASIA CCS)

'23 IEEE Symposium on Security and Privacy (Oakland)

'19, '21, '22, '24, '25, '26 ACM Internet Measurement Conference (IMC)

'25 ACM Internet Measurement Conference (IMC) Student Workshop

'23, '24, '25, '26 The DNS Operations, Analysis, and Research Center (DNS-OARC)

'22, '24, '25, '26 ACM Conference on emerging Networking EXperiments and Technologies (CoNEXT)

'19, '20, '21, '22, '23, '24, '25, '26 The Passive and Active Measurement (PAM)

'20, '21, '24, '25 The ACM/IRTF Applied Networking Research Workshop (ANRW)

'23, '24, '25 ACM Asian Conference on Internet Engineering (AINTEC)

'23 IEEE SecWeb

'23 IEEE MetaCom Workshop

'22, '23 ACM Conference on SIGMETRICS/Performance

'25 ACM Conference on SIGMETRICS/Performance - External Reviewer

'21, '22 IEEE International Conference on Network Protocols (ICNP)

'21 ACM Conference on Computer and Communications Security (CCS) – External Reviewer

'20 The Network Traffic Measurement and Analysis Conference (TMA)

'19 International Conference on Social Informatics (SocInfo)

'17, '18, '19 International AAAI Conference on Web and Social Media (ICWSM)

'17 IEEE International Conference on Ubiquitous Computing and Communications (IUCC)

'14 Poster Selection Committee of ACM/IEEE Symposium on Architectures for Networking and Communications Systems (ANCS)
Shadow Program Committee Member.

'18 ACM Internet Measurement Conference (IMC)

'17 ACM Internet Measurement Conference (IMC)
Artifact Evaluation Member.

'20 International Conference on emerging Networking EXperiments and Technologies (CoNEXT)
Web Chair.

'21, '22 ACM SIGCOMM

'21 ACM Internet Measurement Conference (PAM)

'19 ACM Internet Measurement Conference (IMC)
Travel Grant Chair.

'22 ACM Internet Measurement Conference (IMC)

Registration Chair.

'18 ACM Internet Measurement Conference (IMC)

Journal Reviewer.

'19, '22, '25 IEEE Transactions on Dependable and Secure Computing

'25 Elsevier Computer Communications

'24 Journal of Cybersecurity

'23 Journal of Law and Legislation (KJLL)

'19, '23 ACM Transactions on Privacy and Security

'23 ACM Computing Surveys

'21 Elsevier Computers & Security

'19, '21, '24 IEEE/ACM Transactions on Networking

'16, '19, '20, '24 ACM Computer Communication Review

Proposal Reviewer.

'23 National Science Foundation: CNS:NeTS

'21 The Commonwealth Cyber Initiative (CCI): NextG

Internal and Other Service.

'22 VT Hacks IX Judge

'21 Career Panel: BranchOut, A Youth Community Outreach Program for Local High School Students

'21 VT Hacks IX Judge

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